

CH248 Project Report

Modelling Non-Additive Effects in FUS Phase Separation Using an Extended Flory–Huggins Framework

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Abstract

Liquid–liquid phase separation (LLPS) of intrinsically disordered proteins (IDPs) such as FUS underpins the formation of biomolecular condensates in cells, with important implications for neurodegenerative disease. While the individual effects of salt concentration and macromolecular crowders such as polyethylene glycol (PEG) on LLPS have been studied, their combined, potentially non-additive influence has received far less theoretical attention. This report develops a minimal, three-component extended Flory–Huggins (FH) free energy framework incorporating FUS, PEG, and water, where salt enters as an external control parameter through ionic-strength-dependent interaction parameters. The protein–protein interaction parameter χ_{pp} is endowed with Debye–Hückel screening terms to reproduce reentrant phase behaviour. The crowder–protein interaction χ_{pc} contains a salt–crowder coupling term δ whose non-zero value would signal non-additivity. Phase boundaries are computed numerically via spinodal construction. Using synthetic data generated from the model itself, a stepwise fitting procedure is demonstrated that recovers the underlying parameters with good accuracy. Crucially, the coupling parameter δ is shown to be detectable above realistic noise levels, establishing the framework as a practical tool for interrogating experimental phase diagrams. The model successfully captures reentrant condensation with increasing ionic strength and crowder-driven enhancement of phase separation. Limitations, including the homopolymer treatment of FUS and the implicit mean-field description of salt, are discussed alongside directions for future work.

1. Introduction to the Biological System

1.1. Biomolecular Condensates and Liquid–Liquid Phase Separation

Living cells organise biochemical reactions across space and time using compartments. Classical compartmentation relies on lipid membranes, but it has become clear over the past two decades that compartmentation can also arise through liquid–liquid phase separation of proteins and nucleic acids to form biomolecular condensates or membraneless organelles [1]. These condensates include stress granules, P-bodies, and nucleoli, among many others, and they concentrate specific molecular components while excluding others. Their liquid-like rheology allows rapid molecular exchange with the surrounding cytoplasm and nucleus.

The thermodynamic basis of LLPS is a homogeneous mixture becoming unstable with respect to demixing into a protein-rich dense phase and a protein-poor dilute phase. For IDPs this is driven by a delicate balance of attractive interactions like hydrogen bonds, π – π stacking, and electrostatic interactions, against the entropic cost of localising flexible chains [1].

Fused in sarcoma (FUS) is a widely studied example of a phase-separating IDP. It belongs to the FET family of RNA-binding proteins and contains a disordered low-complexity (LC) domain. FUS condensates participate in RNA processing and the formation of stress granules under cellular stress. Mutations in FUS are linked to amyotrophic lateral sclerosis (ALS), where aberrant phase transitions lead to solid-like, pathological aggregates.

1.2. Effect of Salt and Crowders

Cellular environments are highly crowded, with macromolecules occupying 20–30% of the volume, and ion concentrations are tightly regulated. Both factors modulate LLPS in ways that are incompletely understood.

Salt Concentration Electrostatic interactions within FUS LC domains are screened by counterions, so there are fewer electrostatic interactions between them. The system phase separates at low salt, redissolves at intermediate salt, and phase separates again at high salt, which is called reentrant behaviour, and has been observed experimentally [2] and attributed to competing Debye–Hückel screening (which diminishes electrostatic repulsion at low I) and higher-order ion-correlation effects (which enhance attractions at high I) [3].

Molecular Crowding Agents Polyethylene glycol (PEG) is widely used as a model crowder in biophysical studies because it is chemically inert with respect to most proteins. PEG exerts excluded-volume (depletion) forces that effectively increase the chemical potential of dissolved proteins, thereby promoting condensation. Concentration-dependent

effects have been documented [4], with linear dependence of the effective interaction on PEG volume fraction.

Combined effects When salt and crowder are varied simultaneously, it is not known whether their effects simply add together or whether non-trivial coupling between the salt and crowder generates emergent non-additive behaviour. This question motivates the present modelling effort.

2. Mathematical Approach and Previous Work

2.1. Previous Approaches

Theoretical efforts to describe IDP phase separation have proceeded along several parallel tracks, each capturing different aspects of the underlying physics. The earliest and most tractable approach is classical mean-field Flory–Huggins (FH) theory, which treats the protein as a homogeneous polymer and the solvent as a featureless background; while analytically convenient, it averages over all sequence information and cannot distinguish IDPs with different charge patterns or residue compositions [1].

To incorporate sequence specificity while retaining analytical tractability, Lin, Forman-Kay, and Chan introduced a random-phase-approximation (RPA) framework for polyampholyte phase separation, in which electrostatic interactions are treated at the level of two-body charge-density correlations along the chain [5]. This was a significant advance over mean-field theories because RPA embodies chain connectivity, allowing the precise arrangement of charged residues to influence the predicted phase diagram.

At a still more detailed level of description, the stickers-and-spacers model is used, in which IDP residues are partitioned into interaction-competent stickers (aromatic and charged residues driving multivalent contacts) and flexible spacers that modulate chain dimensions but contribute less to the driving force for condensation [1]. This framework maps molecular grammar onto effective thermodynamic parameters and has become the conceptual reference point for relating IDP sequence to phase behaviour.

For the specific problem of ionic-strength dependence, Mondal and Shakhnovich recently showed that incorporating Debye–Hückel ion-correlation terms into a field-theoretic free energy is sufficient to reproduce the reentrant condensation of FUS, where the competition between low-salt ion-mediated attraction and high-salt depletion from the condensate produces a non-monotonic phase boundary as a function of ionic strength [3].

The influence of macromolecular crowders has been treated separately through depletion-interaction theory, originally due to Asakura and Oosawa [6], in which excluded-volume forces exerted by non-adsorbing polymers generate an effective osmotic attraction between

condensing solutes.

Crucially, none of these frameworks has been extended to treat the cross-coupling between ionic strength and crowder concentration, leaving open the question of whether their combined effect on phase separation is strictly additive, which is the gap that the present work addresses.

2.2. Flory–Huggins Theory

The canonical framework for describing polymer solutions is the Flory–Huggins (FH) lattice theory [1]. This is the most basic theory which can be used for modelling phase separating systems. For a binary polymer–solvent system, the dimensionless free energy of mixing is given by

$$f_{\text{FH}} = \frac{\phi_p}{N_p} \ln \phi_p + (1 - \phi_p) \ln(1 - \phi_p) + \chi \phi_p(1 - \phi_p), \quad (1)$$

where ϕ_p is the volume fraction of the polymer, N_p is its degree of polymerisation, and χ is the Flory–Huggins interaction parameter encoding the enthalpic contribution to free energy.

Phase separation occurs when the free energy of mixing becomes concave in ϕ_p over some concentration range, producing spinodal instability. The binodal (coexistence curve) is found by the common-tangent construction: equating the chemical potentials and osmotic pressures of the two coexisting phases.

For IDP systems, the FH framework must be extended in two ways. First, multicomponent descriptions are needed when more than one solute is present [7]. Second, the interaction parameters χ are not universal constants but depend on solution conditions such as ionic strength [3] and crowder concentration [4].

2.3. Analytical and Numerical Solutions

Qian *et al.* [7] derived closed-form expressions for the binodal of the binary FH model, providing a useful benchmark for numerical solvers. For ternary and higher-order systems, analytical progress is limited and numerical methods are required.

2.4. High-Throughput Experimental Platforms

The PhaseScan platform [8] enables combinatorial mapping of LLPS phase boundaries across hundreds of solution conditions using microfluidic droplet arrays. This can generate datasets that span broad ranges of protein, salt, and crowder concentrations simultaneously, creating a natural opportunity for systematic comparison with theoretical models.

A complementary platform is PhaseXplorer, [9], which integrates microfluidics, real-time microscopy, and machine learning into a closed-loop active learning workflow that autonomously selects the most informative compositions for each experimental iteration. Compared to traditional grid-screen approaches, PhaseXplorer can generate four-dimensional phase diagrams over 100 times faster and with over 10,000 times less material, making it a particularly attractive platform for mapping the high-dimensional condition space.

2.5. Motivation for This Work

While models for salt-dependent [3] and crowder-dependent [4] LLPS have been developed independently, no minimal theoretical framework has been proposed that allows one to test quantitatively whether their combined effect is additive. The present work fills this gap by introducing a coupling term δ into the Flory–Huggins interaction parameters and assessing whether it is detectable from realistic data.

3. Model Formulation and Contributions

An extended Flory–Huggins model with concentration-dependent interaction parameters is used to model a ternary protein-crowder-salt system, where salt is implicitly incorporated via its ionic strength. A cross-interaction parameter between the crowder and salt is introduced.

3.1. Three-Component Extended Flory–Huggins Framework

The system is modelled as a ternary mixture of FUS (p), PEG (c), and water (s). Denoting the volume fractions by ϕ_p , ϕ_c , and $\phi_s = 1 - \phi_p - \phi_c$, the total dimensionless free energy density is

$$f = \frac{\phi_p}{N_p} \ln \phi_p + \frac{\phi_c}{N_c} \ln \phi_c + (1 - \phi_p - \phi_c) \ln(1 - \phi_p - \phi_c) + \chi_{pp}(I) \phi_p^2 + \chi_{pc}(\phi_c, I) \phi_p \phi_c, \quad (2)$$

where the first three terms are the translational entropy contributions and the next two terms encode pairwise interactions. The exact form of the interaction parameters are described below.

3.2. Ionic-Strength-Dependent Interaction Parameters

Salt is incorporated as an external control parameter through the ionic strength I . The protein–protein interaction parameter is written as

$$\chi_{pp}(I) = \chi_{pp}^0 - A\sqrt{I} + BI, \quad (3)$$

where χ_{pp}^0 is the bare protein–protein interaction in the absence of salt. The term $-A\sqrt{I}$ captures Debye–Hückel screening of electrostatic repulsion, which reduces χ_{pp} (makes the protein effectively less soluble) at low I . The term $+BI$ captures higher-order ion–correlation effects that enhance protein–protein attraction at high I [3]. This form is the simplest that can produce reentrant phase behaviour.

The crowder–protein interaction parameter is

$$\chi_{pc}(\phi_c, I) = K_1 + K_2\phi_c + \delta I, \quad (4)$$

where K_1 and K_2 describe concentration-dependent depletion attraction of PEG on FUS, and δ is the salt–crowder coupling term. A non-zero δ implies that the influence of PEG on phase separation is not simply additive with that of salt; the two perturbations interact. The total interaction free energy density is therefore

$$f_{\text{int}} = \chi_{pp}(I) \phi_p^2 + \chi_{pc}(\phi_c, I) \phi_p \phi_c. \quad (5)$$

This model is *minimal* in the sense that it introduces only six free parameters ($\chi_{pp}^0, A, B, K_1, K_2, \delta$) while capturing the key physics of both salt-driven and crowder-driven condensation and their potential coupling.

3.3. Phase Diagram Construction

The phase diagram of the ternary system is characterised by two boundaries: the binodal, which marks the limit of thermodynamic coexistence, and the spinodal, which marks the limit of metastability. The spinodal is determined by the condition that the Hessian matrix of f with respect to ϕ_p and ϕ_c becomes singular,

$$\det \begin{pmatrix} \partial^2 f / \partial \phi_p^2 & \partial^2 f / \partial \phi_p \partial \phi_c \\ \partial^2 f / \partial \phi_p \partial \phi_c & \partial^2 f / \partial \phi_c^2 \end{pmatrix} = 0, \quad (6)$$

inside which the homogeneous mixture is unstable to infinitesimal fluctuations and phase separates spontaneously.

The binodal lies outside the spinodal and encloses the region of metastability; it is found from the coexistence conditions requiring that the chemical potential and osmotic pressure of each species be equal in the coexisting dilute (α) and dense (β) phases:

$$\mu_i^\alpha = \mu_i^\beta, \quad \Pi^\alpha = \Pi^\beta, \quad i \in \{p, c, w\}, \quad (7)$$

where $\mu_i = \partial f / \partial \phi_i$ and the osmotic pressure is $\Pi = -f + \sum_i \phi_i (\partial f / \partial \phi_i)$. Because volume

fractions are not independent ($\phi_w = 1 - \phi_p - \phi_c$), there are effectively two independent chemical potential conditions and one osmotic pressure condition, giving a closed system of equations for the four unknowns ($\phi_p^\alpha, \phi_c^\alpha, \phi_p^\beta, \phi_c^\beta$). These are solved numerically using a Newton–Raphson scheme initialised near the spinodal boundary and the procedure is repeated for nearby points to trace the binodal curve.

4. Results and Discussion

4.1. Synthetic Data Generation

To validate the numerical implementation and fitting procedure, synthetic data were generated by evaluating the spinodal ($\det(H)$) at known parameter values in two stages, mirroring the structure of the fitting algorithm. Points where the spinodal condition was negative were assigned a value of 1, indicating phase separation, while those with positive values were assigned 0, indicating no phase separation. Label noise was introduced to mimic experimental uncertainty in the data generation.

First, ϕ_c was fixed at zero and ϕ_p versus I data were generated. Subsequently, ϕ_p versus ϕ_c data were generated at three fixed values of I . This data structure was chosen to reflect the types of datasets commonly available in the literature.

K_2 was found to have a negligible effect on the spinodal and was therefore fixed to zero in all subsequent analyses. It was excluded from the fitting procedure to improve the robustness of parameter estimation for the remaining variables.

The resulting phase diagrams confirm that the model reproduces two key experimental observations:

1. **Reentrant condensation with salt.** At fixed ϕ_c , the protein-rich phase first appears (low I , Debye–Hückel term dominates), then disappears at intermediate I , and re-emerges at high I as the BI term dominates.
2. **Crowder-driven phase separation.** Increasing ϕ_c shifts the spinodal to lower protein concentrations, expanding the two-phase region. This reflects the depletion-induced effective attraction encoded in χ_{pc} .

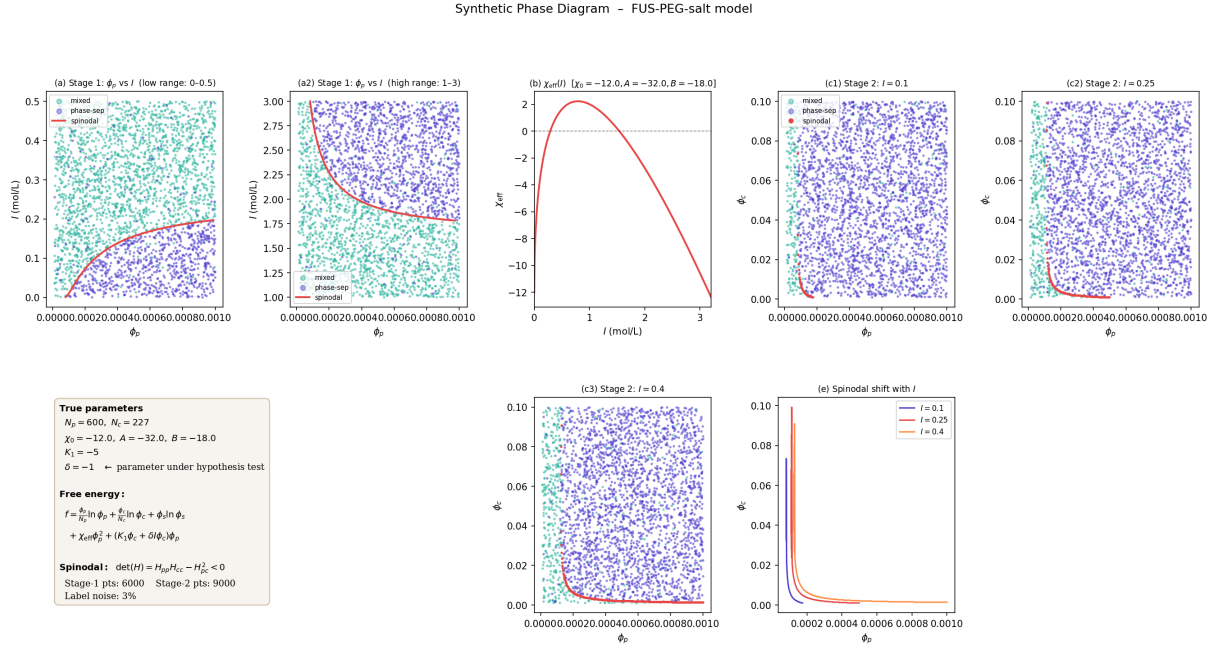


Figure 1: **Synthetically generated phase diagrams.** The summary table gives the parameters used to generate this dataset. A total of 15000 points were generated with a 3% label noise. a) ϕ_p versus I phase diagrams at low I (a1) and high I (a2) range. Phase separation in the high I range is referred to as reentrant behaviour. b) Dependence of χ_{pp} on I . Explains reentrant behaviour, as χ_{pp} is observed to increase and then decrease. Higher χ_{pp} suppresses phase separation. c1, c2, c3) ϕ_p versus ϕ_c phase diagrams at three values of I : 0.1, 0.25, and 0.4 M respectively. Phase boundary shifts to low ϕ_c on addition of crowder, showing that the crowder drives phase separation. d) Shift in spinodal curve with changing I . At the low values of I considered here, increasing I suppresses phase separation, demonstrated by a shift of the spinodal to higher values.

4.2. Stepwise Parameter Fitting

A key contribution of this work is the demonstration that the five remaining model parameters can be recovered in a sequential, identifiable fashion. The fitting procedure proceeds as follows:

1. **Salt-only data** ($\phi_c = 0$) With PEG absent, $\phi_c = 0$ and the free energy reduces to a binary FH model with $\chi_{pp}(I)$ as in Eq. (3). Fitting the ϕ_p versus I spinodal curve recovers χ_{pp}^0 , A and B .
2. **PEG data at varying I** Fitting the ϕ_p versus ϕ_c spinodal curve at different values of I recovers the remaining parameters K_1 and δ . These parameters cannot be differentiated if multiple I values are not considered.

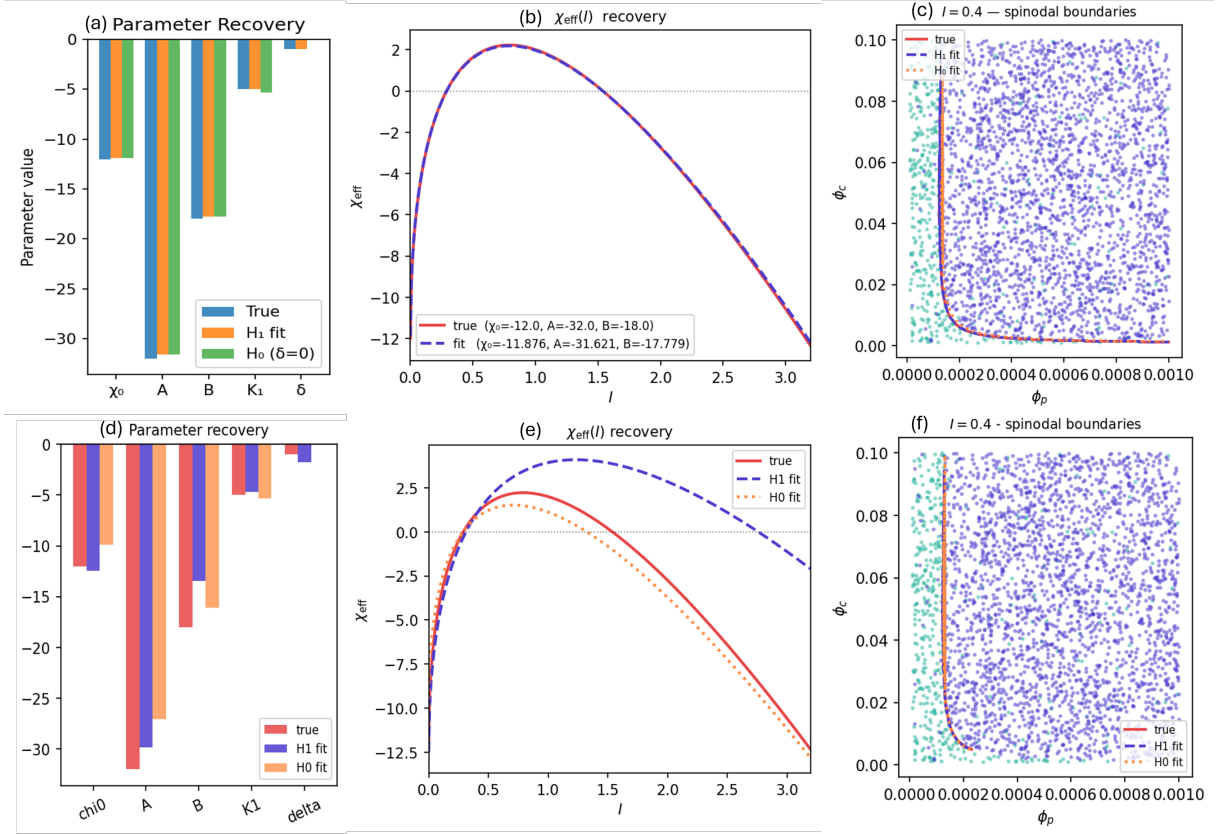


Figure 2: Parameter recovery and comparison of fitting algorithms. Comparing the results of both fitting algorithms: sequential (a-c) and joint (d-f). a) Comparison between the fitted value and true value of the model parameters. H_1 refers to the complete model while H_0 refers to the model where δ is constrained to be 0. χ_{pp}^0 , A , and B are equal for H_0 and H_1 for the sequential fit, because they are fitted in the first step which is independent of δ . The sequential model shows good parameter recovery, but only in presence of the high range I data. b) Recovery of χ_{pp} , shown as a function of I . The fitted values are in excellent agreement with the true values. c) The fitted spinodal boundary also agrees with the true boundary. The H_0 curve is shifted slightly to the right relative to the true spinodal boundary. d) The fitted values of B and δ deviate significantly from their true values in case of joint fitting. e) The deviation of B is reflected by the plot of χ_{pp} versus I . The deviation in B may be due to low sampling along the I axis. f) The spinodal boundary obtained from fitting does not differ much from the true boundary. This suggests that there may be identifiability problems encountered between B and δ during the joint fitting. This is solved by the sequential fitting since it fits them in different steps.

Parameter recovery from synthetic data showed good accuracy for the parameter set tested, with relative errors below 10% for χ_{pp}^0 , A , B , K_1 , and δ under typical noise levels (Figure 2). This confirms that the sequential fitting protocol is consistent and not strongly ill-posed. However, it must be noted that B could be reliably fit only in the presence of the data in the reentrant salt region, otherwise it tended to blow up to strongly negative values.

A joint fitting procedure was also attempted, in which the five parameters $(\chi_{pp}^0, A, B, K_1, \delta)$ were optimised simultaneously against the dataset used in step 2 of the sequential fit, since all three parameters are varied in that dataset. It proved considerably less robust than the sequential approach.

Table 1: Fitted vs. true parameter values under sequential and joint fitting, with relative error $|(\hat{\theta} - \theta)/\theta| \times 100\%$.

Parameter	True value	Sequential fit		Joint fit	
		Fitted	Rel. error (%)	Fitted	Rel. error (%)
χ_{pp}^0	-12.0	-11.8759	1.03	-12.4818	4.02
A	-32.0	-31.6206	1.19	-29.8667	6.67
B	-18.0	-17.7790	1.23	-13.4553	25.25
K_1	-5.0	-5.0043	0.09	-4.6952	6.10
δ	-1.0	-0.9917	0.83	-1.7745	77.45

Parameter recovery errors for B and δ in particular increased significantly (upto 80%) relative to the sequential case, due to lower variation along the I axis, and because only low I values were considered. From Figure 2f, it is seen that the fitted boundary does not differ significantly from the true boundary. This suggests that there is an identifiability problem between B and δ , since both these terms have a quadratic dependence on ϕ and linear dependence on I . This dataset most closely resembles the data already existing in literature, but unfortunately this joint fitting strategy cannot be reliably used for fitting experimental data.

4.3. Detectability of the Coupling Parameter δ

The central scientific question of this project is whether the non-additive coupling term δ can be reliably distinguished from zero given realistic experimental noise.

To address this, two models were fit using the synthetic dataset by sequential fitting: one where $\delta = 0$, and one where δ is allowed to vary freely. The null hypothesis was taken as $\delta = 0$ and the p-value was found to be much lesser than 0.001 for the likelihood ratio test (LRT), indicating that it is significantly nonzero. Other goodness of fit parameters and hypothesis tests, like AIC and BIC tests were performed and gave consistent results. Hence, a nonzero δ can be detected using this framework.

Table 2: Likelihood ratio test results for $H_0 : \delta = 0$ vs. $H_1 : \delta \neq 0$ under sequential fitting.

	Sequential fit
LRT statistic Λ	1252.196
p -value	2.7664×10^{-274}
$\Delta\text{AIC } (H_0 - H_1)$	1250.20
$\Delta\text{BIC } (H_0 - H_1)$	1243.09
95% CI for δ	$[-1.0000, -0.9492]$
Fitted δ	-0.9917
True δ	-1.0
Decision	Reject H_0

5. Conclusion

5.1. Summary

This project developed and analysed a minimal extended Flory–Huggins framework for the three-component FUS–PEG–water system in which salt acts as an external control parameter via ionic-strength-dependent interaction parameters. The model successfully captures reentrant condensation with increasing salt concentration and crowder-driven expansion of the two-phase region—two well-documented experimental phenomena. A salt–crowder coupling term δ was introduced as the key parameter encoding potential non-additive effects.

The main quantitative contribution is the sequential fitting protocol and the accompanying detectability analysis. By demonstrating that (i) the five model parameters can be recovered in a sequential, identifiable fashion from synthetic data, and (ii) the coupling parameter δ is distinguishable from zero, the framework provides a concrete route from experimental phase diagrams to a quantitative verdict on non-additivity. The model being based in Flory–Huggins theory makes it simple and easily interpretable, while capturing novel interactions that have not been explored before. The ionic-strength parameterisation of χ_{pp} is phenomenological but grounded in Debye–Hückel theory, which lends it some mechanistic credibility.

At the same time, the model does have significant limitations since it is a minimal model based on the mean-field Flory–Huggins theory. The homopolymer treatment of FUS erases all sequence-level information: the patchy, multivalent character of the LC domain, with its distinct aromatic and charged residues, is reduced to a single effective χ parameter. Salt is treated implicitly (as a mean-field ionic strength) rather than as an explicit molecular species; this precludes modelling salt partitioning between phases, which can be experimentally significant. Finally, the model does not distinguish between different types of crowders: PEG is treated purely as a depletion agent, whereas some

crowders can form specific interactions with IDPs.

5.2. Future Work

Several natural extensions of this work would increase its biological relevance:

1. **Comparison with PhaseScan data.** The immediate next step is to apply the fitting procedure to the high-throughput experimental data from Ref. [8] and extract quantitative estimates of δ for FUS. Confidence intervals should be reported to make the non-additivity claim statistically rigorous.
2. **Explicit salt treatment.** Incorporating salt as a fourth component with its own translational entropy and electrostatic free energy (e.g., via Poisson–Boltzmann theory) would allow modelling of salt partitioning and ion-specific effects (Hofmeister series), which the current model ignores.
3. **Sequence-aware models.** Stickers-and-spacers models [1], in which specific residues drive multivalent attractive interactions, could be combined with the FH framework to connect the macroscopic χ parameters to molecular sequences, enabling predictions for FUS mutants relevant to disease.
4. **Alternative crowders.** Comparing δ values for PEG, Ficoll, and dextran would test the generality of the non-additivity effect and potentially discriminate between entropy-driven (depletion) and enthalpically driven crowder–protein interactions.

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